暖念纪念场景模拟软件 V1.0 · 源程序（前30页，共60页） 说明：每页不少于50行；页眉为软件名称与版本号，右下角为页码。
/* ===== 源文件：src/App.tsx ===== */ /**

- @file App.tsx
- @description 应用外壳（App Shell）：只负责视图路由、登录态持久化与全局弹窗宿主。

*

- 具体业务页面（馆空间）位于 src/ui/space，登录页位于 src/ui/auth，
- 本文件不承载任何场景陈设、摆件或弹窗内容。

*/ import React, { useEffect, useState } from 'react'; import { AuthPage } from './ui/auth'; import { HallSpacePage } from './ui/space'; import { openAgreement } from './ui/agreement'; import { useViewRouter } from './ui/shell/viewRouter'; import type { UserProfile } from './types';

const USER_STORAGE_KEY = 'memery_current_user';

const readStoredUser = (): UserProfile | null => { try { const saved = localStorage.getItem(USER_STORAGE_KEY); return saved ? (JSON.parse(saved) as UserProfile) : null; } catch { return null; } };

const App: React.FC = () => { const { view, navigate } = useViewRouter(); const [currentUser, setCurrentUser] = useState<UserProfile | null>(readStoredUser);

// 登录态持久化集中在外壳，页面不直接接触存储 useEffect(() => { try { if (currentUser) { localStorage.setItem(USER_STORAGE_KEY, JSON.stringify(currentUser)); } else { localStorage.removeItem(USER_STORAGE_KEY); } } catch (err) { console.error(err); } }, [currentUser]);

if (view === 'auth') { return (<AuthPage onBack={() => navigate('hall')} onSuccess={(user) => {

setCurrentUser(user); navigate('hall'); }} onOpenAgreement={openAgreement} />); }

return (<HallSpacePage currentUser={currentUser} onRequestAuth={() => navigate('auth')} onUserChange={setCurrentUser} />); };

export default App;

/* ===== 源文件：src/data/lore/flowers.ts ===== */ /**

- @file flowers.ts
- @description 供花典故。id 与 FlowerType 一致，可直接按花材 id 查档。

*/ import type { RelicLore } from './types';

export const FLOWER_LORE: RelicLore[] = [{ id: 'peonies', category: 'flower', name: '牡丹', poeticTitle: '国色天香', dynasty: '唐', origin: '唐 · 李肇《唐国史补》', allusion: '唐人最尚牡丹，长安春暮花时，士庶车马若狂，一株有直数万钱者。《唐国史补》记：「京城贵游尚牡丹三十余年矣，每春暮车马若狂。」故牡丹为唐世花王，非虚誉。', poem: { text: '唯有牡丹真国色，花开时节动京城。', title: '赏牡丹', author: '唐 · 刘禹锡', }, ritual: '供花以双数为敬，花头朝神位；牡丹用于春祭，取富贵圆满之意。', authenticity: 'canon', related: ['vase'], }, { id: 'lotus', category: 'flower', name: '莲',

poeticTitle: '净土清芬', dynasty: '南北朝', origin: '宋 · 周敦颐《爱莲说》', allusion: '莲随佛教东传而入中土供奉，佛经以莲花出淤泥而不染喻清净法身，造像与藻井皆以莲为座。宋儒周敦颐作《爱莲说》，遂以莲比君子。', poem: { text: '出淤泥而不染，濯清涟而不妖。', title: '爱莲说', author: '宋 · 周敦颐', }, ritual: '佛前供莲最古，宜清水养，忌萎凋不换。', authenticity: 'canon', related: ['pattern_lotus'], }, { id: 'chrysanthemum', category: 'flower', name: '菊', poeticTitle: '晚节凌霜', dynasty: '晋', origin: '晋 · 陶渊明《饮酒》、南朝 · 吴均《续齐谐记》', allusion: '陶渊明采菊东篱，菊遂为隐逸与晚节之象。重九佩茱萸、饮菊酒之俗，见《续齐谐记》所载桓景登高避灾事。', poem: { text: '采菊东篱下，悠然见南山。', title: '饮酒 · 其五', author: '晋 · 陶渊明', }, ritual: '菊耐霜不落，喻人之晚节，秋祭与重阳扫墓多用之。', authenticity: 'canon', }, { id: 'plum_blossoms', category: 'flower', name: '梅', poeticTitle: '暗香疏影', dynasty: '宋', origin: '《宋史 · 隐逸传》、宋 · 林逋《山园小梅》', allusion: '宋人林逋隐居孤山，植梅畜鹤，世谓「梅妻鹤子」，咏梅遂成一代风气。梅开百花之先，凛寒独放，与松、竹并为岁寒三友。', poem: { text: '疏影横斜水清浅，暗香浮动月黄昏。', title: '山园小梅', author: '宋 · 林逋', }, ritual: '冬春之交供梅，取凌寒守志之意，宜疏不宜密。', authenticity: 'canon', }, { id: 'orchids',

category: 'flower', name: '兰', poeticTitle: '空谷幽香', dynasty: '先秦', origin: '《孔子家语 · 在厄》、战国 · 屈原《离骚》', allusion: '孔子厄于陈蔡，谓「芝兰生于深林，不以无人而不芳」，以兰喻君子慎独之德。屈原则「纫秋兰以为佩」，兰为香祖，古人以祓除不祥。', poem: { text: '芝兰生于深林，不以无人而不芳。', title: '孔子家语 · 在厄', author: '先秦 · 孔门弟子记', }, ritual: '清明插兰、佩兰，古有祓除之礼；供兰宜单株清供，忌浓香夺主。', authenticity: 'canon', }, { id: 'narcissus', category: 'flower', name: '水仙', poeticTitle: '凌波仙子', dynasty: '宋', origin: '宋 · 黄庭坚《王充道送水仙花五十枝》', allusion: '水仙宋时已入文人清供，黄庭坚以洛神拟之，名曰「凌波仙子」。漳州水仙自明清为贡品，岁朝案头必有。', poem: { text: '凌波仙子生尘袜，水上轻盈步微月。', title: '王充道送水仙花五十枝', author: '宋 · 黄庭坚', }, ritual: '岁朝清供首选，清水养、素器盛，取洁净迎新之意。', authenticity: 'canon', related: ['vase'], }, { id: 'roses', category: 'flower', name: '玫瑰', poeticTitle: '浅深两色', dynasty: '宋', origin: '宋 · 杨万里《红玫瑰》', allusion: '宋人已辨玫瑰非月季、非蔷薇，杨万里诗云「不与蔷薇谱牒通」。古之玫瑰入香、入药、入馔为其要；今以切花成束供奉，乃近世西俗。', poem: { text: '非关月季姓名同，不与蔷薇谱牒通。', title: '红玫瑰', author: '宋 · 杨万里', }, note: '古典玫瑰重香与色，不以束花献祭；今之「献一束玫瑰」通行于近代，属礼俗流变。', authenticity: 'canon', },

{ id: 'lilies', category: 'flower', name: '百合', poeticTitle: '百年好合', dynasty: '宋', origin: '宋 · 罗愿《尔雅翼》', allusion: '「百合」之名取其鳞茎众瓣合抱，《尔雅翼》谓其「根以众瓣合成」。民间因「百」「合」二字，转出「百年好合」吉语。', note: '今世以百合寄哀思、表纯洁，源自西方花语；与中国固有「百年好合」的婚嫁吉语并非一源。', ritual: '白百合近世多用于追思场合，宜素器、单数清供。', authenticity: 'folk', }, { id: 'hydrangeas', category: 'flower', name: '绣球', poeticTitle: '八仙聚瑞', dynasty: '宋', origin: '宋人「聚八仙」之名', allusion: '木本绣球宋已有之，宋人谓之「聚八仙」，以其花序八朵攒聚如仙侣环坐。明清始通行「绣球」之名。', note: '「聚八仙」与扬州「琼花」古来并称，二者形近而实异，宋人已辨之。', ritual: '花球圆满，取团圆聚会之意，宜置于供案两侧。', authenticity: 'folk', }, { id: 'tulips', category: 'flower', name: '郁金香', poeticTitle: '琥珀浮光', dynasty: '清', origin: '唐 · 李白《客中行》（辨误）', allusion: '今之郁金香为荷兰球根花卉，清代以降渐入中土，二十世纪始大规模引种。古书中「郁金香」另是一物，乃姜黄属香草，用以和酒、染衣。', poem: { text: '兰陵美酒郁金香，玉碗盛来琥珀光。', title: '客中行', author: '唐 · 李白', }, note: '李白诗中「郁金香」指浸了郁金香草的酒香，与今日荷兰郁金香花同名而异物，不可混为一谈。', authenticity: 'modern', }, { id: 'sunflowers', category: 'flower', name: '向日葵', poeticTitle: '倾心向阳', dynasty: '明末', origin: '明 · 王象晋《群芳谱》（一名迎阳花）',

allusion: '向日葵原产美洲，明后期经南洋传入，古称「丈菊」「西番菊」「迎阳花」，王象晋《群芳谱》已著录。', note: '古诗文中单言「葵」者多指葵菜，如「惟有葵花向日倾」，与今之向日葵非一物。', ritual: '取倾心向阳之意，民间用以喻子孙心念先人不改。', authenticity: 'modern', }, { id: 'carnations', category: 'flower', name: '康乃馨', poeticTitle: '寸草春晖', dynasty: '近现代', origin: '欧美石竹属栽培种', allusion: '康乃馨为石竹属洋花，清末民国随西礼传入，西方以之表母爱与追念，近世遂成祭扫常花。', note: '中国古有石竹、瞿麦，然与今之康乃馨（香石竹）并非一物，古亦无以此献祭之制。', ritual: '近世追思用花，宜素色，与白菊同供。', authenticity: 'modern', related: ['chrysanthemum'], }, { id: 'calla_lilies', category: 'flower', name: '马蹄莲', poeticTitle: '素心一瓣', dynasty: '近现代', origin: '南部非洲原产，清末传入', allusion: '马蹄莲原产非洲湿地，清代末期作为观赏花卉引入，因其佛焰苞洁白如绢，近世多用于素净清供。', note: '古无马蹄莲，亦无相关诗咏，其供祭用法出自近世园艺礼俗。', authenticity: 'modern', }, { id: 'lily_of_valley', category: 'flower', name: '铃兰', poeticTitle: '五月清铃', dynasty: '近现代', origin: '欧洲温带原产', allusion: '铃兰为欧洲林下小花，西方称「五月铃」，视为归春与纯洁之象，近现代引入中土。', note: '中文「铃兰」之名晚起，古籍所载铃兰多另指他物，与欧洲铃兰传统无涉。', authenticity: 'modern', }, { id: 'daisies', category: 'flower', name: '雏菊', poeticTitle: '天真烂漫', dynasty: '近现代',

origin: '欧洲原产，近世引入', allusion: '雏菊为欧洲常见野花，近代随园艺传入，花小丛生，取天真、追念之意。', note: '古无雏菊之名，中国古籍中的「菊」皆指菊科菊属，非此西洋小菊。', authenticity: 'modern', related: ['chrysanthemum'], },];

/* ===== 源文件：src/data/lore/fruits.ts ===== */ /**

- @file fruits.ts
- @description 供果典故。id 与 FruitType 一致，可直接按果品 id 查档。

*/ import type { RelicLore } from './types';

export const FRUIT_LORE: RelicLore[] = [{ id: 'peaches', category: 'fruit', name: '桃', poeticTitle: '福寿绵长', dynasty: '先秦', origin: '《诗经 · 周南 · 桃夭》、旧题汉 · 班固《汉武故事》', allusion: '《诗经》以「桃之夭夭，灼灼其华」咏嫁娶之盛；汉魏方士又传西王母蟠桃三千年一熟，东方朔窃桃之说，桃遂为寿果之宗。', poem: { text: '桃之夭夭，灼灼其华。', title: '诗经 · 周南 · 桃夭', author: '先秦 · 无名氏', }, ritual: '祝寿、祭祖皆用寿桃，取「桃」谐「逃」以避邪，又寓长寿。', authenticity: 'canon', }, { id: 'oranges', category: 'fruit', name: '橘', poeticTitle: '大吉大利', dynasty: '先秦', origin: '《尚书 · 禹贡》、战国 · 屈原《楚辞 · 九章 · 橘颂》', allusion: '《禹贡》载扬州「厥包橘柚锡贡」，橘为三代贡品。屈子作《橘颂》：「受命不迁，生南国兮」，以橘之不易其地，喻士之守志不移。', poem: { text: '后皇嘉树，橘徕服兮。受命不迁，生南国兮。', title: '九章 · 橘颂', author: '战国 · 屈原', }, ritual: '橘谐「吉」，新春与祭祖皆供，取吉星高照；南方供橘多带枝叶，示根本不改。', authenticity: 'canon',

}, { id: 'pomelos', category: 'fruit', name: '柚', poeticTitle: '福佑团圆', dynasty: '先秦', origin: '《尚书 · 禹贡》', allusion: '柚与橘并载《禹贡》，同为荆州、扬州之贡。柚大而成双蒂者尤贵，民间谐「柚」为「佑」「有」，谓有神护佑。', ritual: '柚子个大圆满，供于案前取团圆与庇佑之意，闽粤尤重。', authenticity: 'canon', related: ['oranges'], }, { id: 'grapes', category: 'fruit', name: '葡萄', poeticTitle: '珠圆玉润', dynasty: '汉', origin: '汉 · 司马迁《史记 · 大宛列传》', allusion: '葡萄自张骞通西域后传入，《史记 · 大宛列传》记「宛左右以蒲陶为酒，富人藏酒至万余石」，汉家始得其种于内地。', poem: { text: '葡萄美酒夜光杯，欲饮琵琶马上催。', title: '凉州词', author: '唐 · 王翰', }, ritual: '葡萄成串多子，供之以寓子孙繁衍、绵延不绝。', authenticity: 'canon', }, { id: 'persimmons', category: 'fruit', name: '柿', poeticTitle: '事事如意', dynasty: '唐', origin: '唐 · 段成式《酉阳杂俎》', allusion: '《酉阳杂俎》列柿有七绝：一多寿、二多阴、三无鸟巢、四无虫蠹、五霜叶可玩、六嘉宾、七落叶肥大可以临书。', note: '民间谐「柿」为「事」，故年节供柿取「事事如意」，此为谐音吉语，非古礼明文。', ritual: '秋祭供柿，成双摆放，取事事顺遂。', authenticity: 'canon', }, { id: 'plums', category: 'fruit', name: '李', poeticTitle: '投桃报李', dynasty: '先秦',

origin: '《诗经·大雅·抑》、汉·司马迁《史记·李将军列传》', allusion: '《诗》曰「投我以桃，报之以李」，李为往还相报之礼果；太史公赞李广引谚「桃李不言，下自成蹊」，李又喻实德感人。', poem: { text: '投我以桃，报之以李。', title: '诗经·大雅·抑', author: '先秦·无名氏', }, ritual: '桃李并供，喻先人遗泽、后人相报。', authenticity: 'canon', related: ['peaches'], }, { id: 'longans', category: 'fruit', name: '龙眼', poeticTitle: '益智归元', dynasty: '汉晋', origin: '《神农本草经》、晋·嵇含《南方草木状》', allusion: '《神农本草经》已录龙眼，一名益智，谓其「主五脏邪气，安志厌食」。《南方草木状》记岭南龙眼树如荔枝而枝叶稍小。', ritual: '岭南供果常品，取团圆（圆果）与益智（思慕明达）之意。', authenticity: 'canon', }, { id: 'citrons', category: 'fruit', name: '佛手柑', poeticTitle: '福寿在手', dynasty: '明清', origin: '明·李时珍《本草纲目·果部》', allusion: '香橼之变种，果实裂如手指，故名佛手。《本草纲目》谓枸橼「气味辛甘，无毒」，明清士人置之案头清供，取其香与形。', note: '「佛手」谐「福寿」，为明清以降的民间吉语，清宫岁朝图中常见，非古祭礼必用之果。', ritual: '岁朝清供与祭祖并用，取其香不取其食，供毕可留香数月。', authenticity: 'folk', }, { id: 'kumquats', category: 'fruit', name: '金橘', poeticTitle: '金玉满堂', dynasty: '宋', origin: '宋·欧阳修《归田录》', allusion: '金橘宋时已为珍果，欧阳修《归田录》记其香清味美、远致为难。金橘可连皮食，色如金弹，遂成岁朝吉果。', ritual: '谐「金」「吉」，新春与祭祖并供，取金玉满堂。', authenticity: 'canon', related: ['oranges'], },

{ id: 'bananas', category: 'fruit', name: '香蕉', poeticTitle: '长青如扇', dynasty: '晋', origin: '晋 · 嵇含《南方草木状》', allusion: '古称甘蕉，《南方草木状》记其「叶大如席，其茎解散如丝」，南方以为常果。今食用香蕉为近世改良品种。', note: '古籍所载甘蕉与今日栽培香蕉品种有别；香蕉作为案前供果，主要通行于岭南民间。', ritual: '成梳供奉，取「招财进宝」「长青不老」之意，为岭南祭俗。', authenticity: 'folk', }, { id: 'coconuts', category: 'fruit', name: '椰子', poeticTitle: '一瓢清甘', dynasty: '晋', origin: '晋 · 嵇含《南方草木状》', allusion: '《南方草木状》记椰树「叶如栟榈，高六七丈」，其实大如寒瓜，中有白肤、有浆，饮之如蜜。汉赋中「胥邪」一说即指椰。', note: '椰非中原旧产，古为岭南、海外之果，入供属地方礼俗。', ritual: '椰实浑圆、浆汁清甘，取清心明志之意，多见于岭南与南洋华侨祭俗。', authenticity: 'canon', }, { id: 'watermelons', category: 'fruit', name: '西瓜', poeticTitle: '清凉涤暑', dynasty: '五代', origin: '五代 · 胡峤《陷虏记》', allusion: '胡峤入契丹，记其地「西瓜，大如中国冬瓜而味甘」，为西瓜入华的最早明确记载；元明以后南北遍植。', note: '西瓜之入中国，一说经西域、一说随契丹南传，五代为其确证之时，非先秦旧有。', ritual: '暑月祭祖供西瓜，取清凉荐新之意。', authenticity: 'canon', }, { id: 'cucumbers', category: 'fruit', name: '黄瓜', poeticTitle: '青玉长条', dynasty: '汉', origin: '晋 · 陆翙《邺中记》', allusion: '黄瓜本名胡瓜，张骞通西域后传入。后赵石勒讳「胡」字，凡胡物皆改名，《邺中记》载「胡瓜曰黄瓜」，其名沿用至今。', note: '供果用瓜，古多用甜瓜（香瓜）；以黄瓜入供，属地方性、近代的变通用法。', authenticity: 'canon',

}, { id: 'pineapples', category: 'fruit', name: '菠萝', poeticTitle: '旺来呈祥', dynasty: '明清', origin: '明末自南洋传入', allusion: '菠萝约明代后期自南洋传入，其形似波罗蜜而得名，闽南语呼「凤梨」，谐「旺来」，遂成吉果。', note: '闽台「凤梨旺来」为方言谐音吉语，属民间礼俗，古无此制。', ritual: '闽台祭祖、迎神多供凤梨，取兴旺到来之意。', authenticity: 'modern', }, { id: 'pears', category: 'fruit', name: '梨', poeticTitle: '让梨知礼', dynasty: '汉', origin: '南朝宋 · 范晔《后汉书 · 孔融传》注引《融家传》', allusion: '孔融四岁与诸兄共食梨，辄引小者，曰「我小儿，法当取小者」，事见《融家传》，为悌让之典。', note: '民间以「梨」谐「离」，祭祖、婚喜多避用梨，此为谐音禁忌而非古籍禁令，收录以备考。', ritual: '供果多避梨；若用，宜成双不分离，或改供他果。', authenticity: 'canon', }, { id: 'apples', category: 'fruit', name: '苹果', poeticTitle: '岁岁平安', dynasty: '明', origin: '明 · 王象晋《群芳谱》', allusion: '古之绵苹果称「柰」，佛经译名「频婆果」另指一物；「苹果」之名通行于明清，《群芳谱》载「苹果出北地，燕赵者尤佳」。', note: '今供案所用红富士为二十世纪日本培育品种。以「苹」谐「平安」是近现代民俗，古礼并无苹果供祭之制。', ritual: '今世祭祖、年节供苹果取平安之意，成双或四枚为敬。', authenticity: 'modern', }, { id: 'dragonfruits', category: 'fruit', name: '火龙果', poeticTitle: '赤鳞献瑞', dynasty: '近现代', origin: '中美洲原产，二十世纪末引入', allusion: '火龙果为仙人掌科量天尺之果，原产中美洲，二十世纪末引入东南亚与中国，因其色红似火、外披鳞片，民间以「火龙」名之。', note: '域外新果，古籍无载，亦无典故诗咏；供之取其色赤喜庆。',

authenticity: 'modern', }, { id: 'mangosteens', category: 'fruit', name: '山竹', poeticTitle: '紫壳冰心', dynasty: '近现代', origin: '马来群岛原产，近世引入', allusion: '山竹原产马来群岛，二十世纪传入中国南方，果壳紫厚、内瓣洁白，热带果品之珍。', note: '域外新果，古籍无载；入供属近世南洋华侨礼俗的延伸。', authenticity: 'modern', },];

/* ===== 源文件：src/data/lore/index.ts ===== */ /**

- @file index.ts
- @description 物件典藏总索引：聚合四类档案，提供查档与场景物件映射。

*/ import type { LoreCategory, RelicLore } from './types'; import { FLOWER_LORE } from './flowers'; import { FRUIT_LORE } from './fruits'; import { VESSEL_LORE } from './vessels'; import { PATTERN_LORE } from './patterns';

export * from './types'; export { FLOWER_LORE, FRUIT_LORE, VESSEL_LORE, PATTERN_LORE };

/** 全量档案 */ export const ALL_LORE: RelicLore[] = [...FLOWER_LORE, ...FRUIT_LORE, ...VESSEL_LORE, ...PATTERN_LORE];

const LORE_MAP: Map<string, RelicLore> = new Map(ALL_LORE.map((item) => [item.id, item]));

/** 按 id 查档；未收录返回 undefined */ export const getRelicLore = (id?: string | null): RelicLore | undefined => id ? LORE_MAP.get(id) : undefined;

/** 是否已收录典故 */ export const hasRelicLore = (id?: string | null): boolean => !!id && LORE_MAP.has(id);

/** 按门类取档（图鉴分页用） */ export const getLoreByCategory = (category: LoreCategory): RelicLore[] => ALL_LORE.filter((item) => item.category === category);

/** 场景物件携带的可变材质信息（花材 / 器型 / 果品） */ export interface LoreSceneItemHint { flowerType?: string; vaseStyle?: string;

fruitType?: string; }

/**

- 把场景物件 id 映射到档案 id。
- 供瓶优先返回所插花材的典故，其次器型，最后落到「供瓶」总条目。

*/ export const resolveSceneItemLoreId = (sceneId: string, item?: LoreSceneItemHint): string | undefined => { if (sceneId === 'censer') return 'censer'; if (sceneId.startsWith('candle')) return 'candle'; if (sceneId === 'tea_cups') return 'tea_cups'; if (sceneId === 'singing_bowl') return 'singing_bowl';

if (sceneId.startsWith('fruit')) { return item?.fruitType || undefined; }

if (sceneId.startsWith('vase') || sceneId.startsWith('offered_flower')) { if (item?.flowerType && hasRelicLore(item.flowerType)) return item.flowerType; if (item?.vaseStyle && hasRelicLore(item.vaseStyle)) return item.vaseStyle; return 'vase'; }

return hasRelicLore(sceneId) ? sceneId : undefined; };

/** 桌布主题 → 其边饰纹样档案 id（无明确纹样者不映射） */ const TABLECLOTH_PATTERN_LORE: Record<string, string> = { autumn_gold: 'pattern_meander', cinnabar_red: 'pattern_wanzi', moonlight_white: 'pattern_ruyi', evergreen_pine: 'pattern_badayun', sandalwood_purple: 'pattern_lianzhu', };

/** 取桌布边饰所用纹样的档案 id */ export const resolveTableclothLoreId = (tableclothStyle?: string | null): string | undefined => tableclothStyle ? TABLECLOTH_PATTERN_LORE[tableclothStyle] : undefined;

/** 关键词检索（名称、诗意名、朝代、典故、诗句） */ export const searchRelicLore = (keyword: string): RelicLore[] => { const kw = keyword.trim().toLowerCase(); if (!kw) return ALL_LORE; return ALL_LORE.filter((item) => [item.name, item.poeticTitle, item.dynasty, item.origin, item.allusion, item.poem?.text, item.poem?.title] .filter(Boolean) .some((field) => (field as string).toLowerCase().includes(kw))); };

/* ===== 源文件：src/data/lore/patterns.ts ===== */ /**

- @file patterns.ts
- @description 纹样典故：回纹、万字、如意、连珠、八达晕、莲纹、蝠纹、宝相花。
- id 以 pattern_ 前缀，与桌布/墙面 borderPatternId、墙面主题对应。

*/ import type { RelicLore } from './types';

export const PATTERN_LORE: RelicLore[] = [{ id: 'pattern_meander', category: 'pattern', name: '回纹', poeticTitle: '连绵不断', dynasty: '商周', origin: '商周青铜器与陶器之云雷纹、回字纹', allusion: '回纹方折回旋，一笔不断，源于商周青铜器上的云雷纹。因其首尾相连、无起无止，民间谓之「富贵不断头」，寓意福泽连绵。', ritual: '多用于案围、织锦与建筑边饰，取久长不绝之意。', authenticity: 'canon', }, { id: 'pattern_wanzi', category: 'pattern', name: '万字纹', poeticTitle: '万德庄严', dynasty: '唐', origin: '佛教瑞相，唐武则天时定读为「万」', allusion: '「卍」本为佛之瑞相，寓吉祥万德。唐武则天长寿年间定此符号读「万」，取万德庄严、吉祥万福之义，遂通行于织绣与装修。', note: '「定读为万」一说流传甚广，具体年份诸书微有出入，此处取通行之说。', ritual: '连缀成「万字不到头」，用于案围与檐角，寓万福无疆。', authenticity: 'canon', }, { id: 'pattern_ruyi', category: 'pattern', name: '如意纹', poeticTitle: '称心如意', dynasty: '魏晋', origin: '源出爪杖（搔杖），明清演为云头纹', allusion: '如意本为背痒时搔之的爪杖，魏晋名士清谈执之，后世渐成礼器与祥物。其首作云头或心形，纹样化后即为如意云头。', ritual: '取「如意」之名，寓事事顺遂，常见于供案织品与器物边饰。', authenticity: 'canon', }, { id: 'pattern_lianzhu', category: 'pattern', name: '连珠纹',

poeticTitle: '珠联璧合', dynasty: '南北朝', origin: '源出波斯萨珊，经丝路东传', allusion: '连珠纹以圆珠相接成圈，其源在波斯萨珊王朝，经丝路传入中土，南北朝至隋唐织锦多见「联珠对鸟」「联珠团窠」之属。', note: '连珠为中外交流之证，非中土原生纹样。', ritual: '取珠联璧合之意，多用于织锦边饰与锦缘。', authenticity: 'canon', }, { id: 'pattern_badayun', category: 'pattern', name: '八达晕', poeticTitle: '四通八达', dynasty: '宋', origin: '宋锦几何纹之一，元 · 费著《蜀锦谱》著录', allusion: '八达晕为宋锦名目之一，以规矩的几何骨架向八方延展，中嵌团花，故名「八达」。元人费著《蜀锦谱》所记宋代蜀锦纹样中即有八答晕。', ritual: '取四通八达、路路皆通之意，宜作案围锦缘。', authenticity: 'canon', }, { id: 'pattern_lotus', category: 'pattern', name: '莲瓣纹', poeticTitle: '莲华化生', dynasty: '南北朝', origin: '佛教东传后的莲瓣装饰（南朝青瓷莲瓣尊一系）', allusion: '莲瓣纹随佛教东传而盛，南朝青瓷仰覆莲瓣尊为其翘楚，后及于石刻、藻井与织品。莲出淤泥不染，为清净之象。', ritual: '佛前供器多用莲瓣纹，取化生净土之意。', authenticity: 'canon', related: ['lotus'], }, { id: 'pattern_bat', category: 'pattern', name: '蝠纹', poeticTitle: '五蝠捧寿', dynasty: '清', origin: '清代吉样纹样，取「蝠」谐「福」', allusion: '清人以「蝠」谐「福」，五蝠拱寿谓「五福捧寿」，出《尚书 · 洪范》五福之说（寿、富、康宁、攸好德、考终命）。', note: '蝠入纹样为明清之事，谐「福」属民间吉语，先秦并无以蝠为纹之制。', ritual: '祝寿与祭祖场合皆用，取福寿双全。', authenticity: 'folk', }, { id: 'pattern_baoxiang', category: 'pattern',

name: '宝相花', poeticTitle: '宝相庄严', dynasty: '唐', origin: '唐代佛教装饰团花', allusion: '宝相花非实有之花，乃唐人以莲、牡丹、石榴诸花合为一体，作对称团花，用于佛帐、藻井与织锦，取其庄严圆满。', note: '宝相花为「理想花」，非某种真实花卉，故不宜按花名索解。', ritual: '多见于供案织品与壁饰，取圆满庄严之意。', authenticity: 'canon', },];

/* ===== 源文件：src/data/lore/types.ts ===== */ /**

- @file types.ts
- @description 「物件典藏」文化元数据协议层。

*

- 每一花、一器、一果、一纹样，都记录其朝代、出处、典故与诗证。
- 关键约定：authenticity 必须如实标注 —— 典籍可考 / 民俗相传 / 近现代，
- 不可把民俗谐音包装成古籍出处，也不可给近代传入的作物硬编朝代。

*/

/** 出处可信度分级 */ export type LoreAuthenticity = 'canon' | 'folk' | 'modern';

/** 物件门类 */ export type LoreCategory = 'flower' | 'fruit' | 'vessel' | 'pattern';

/ **诗句佐证** */ **export interface RelicPoem { / 诗句正文** */ text: string; / **诗题** */ title: string; / 作者（含朝代，如「唐 · 刘禹锡」）** */ author: string; }

/ **单件物件的文化档案** */ **export interface RelicLore { / 唯一 id：花/果沿用业务 id（如 'peonies'、'apples'），器与纹样用独立 id** */ id: string; category: LoreCategory; / **通行名称** */ **name: string; / 四字诗意别名** */ poeticTitle?: string; / **盛行或最早可考的朝代** */ **dynasty?: string; / 出处：文献篇目或实物来源** */

origin?: string; / 典故正文 */ allusion: string; / 诗证（无可靠诗证时留空，不硬凑） */ poem?: RelicPoem; / 祭祀礼俗中的用法与宜忌 */ ritual?: string; / 辨误或补充（同名异物、古义今义之别） */ note?: string; / 出处可信度 */ authenticity: LoreAuthenticity; / 相关物件 id，可在卡片内跳转 */ related?: string[]; }

export const AUTHENTICITY_META: Record< LoreAuthenticity, { label: string; hint: string; color: string; bg: string }

= {

canon: { label: '典籍可考', hint: '有传世文献或出土实物佐证', color: '#0f766e', bg: 'rgba(15,118,110,0.10)', }, folk: { label: '民俗相传', hint: '民间吉语、谐音与岁时礼俗，非古籍出处', color: '#a16207', bg: 'rgba(161,98,7,0.10)', }, modern: { label: '近现代', hint: '域外传入或近世通行，古无此制', color: '#64748b', bg: 'rgba(100,116,139,0.10)', }, };

export const CATEGORY_LABEL: Record<LoreCategory, string> = { flower: '供花', fruit: '供果', vessel: '礼器', pattern: '纹样', };

/** 真实性分级的展示顺序（卡片徽标排序用） */ export const AUTHENTICITY_ORDER: LoreAuthenticity[] = ['canon', 'folk', 'modern'];

/* ===== 源文件：src/data/lore/vessels.ts ===== */

/**

- @file vessels.ts
- @description 礼器典故：香炉、烛台、茶盏、铜磬、供瓶，以及各瓶型的取法谱系。

*

- 注意：本作器型多为当代设计，档案中「取法」指其形制渊源，不代表出土原物。

*/ import type { RelicLore } from './types';

export const VESSEL_LORE: RelicLore[] = [{ id: 'censer', category: 'vessel', name: '香炉', poeticTitle: '博山烟直', dynasty: '汉', origin: '旧题汉 · 刘歆《西京杂记》', allusion: '《西京杂记》记长安巧工丁缓作九层博山香炉，镂以奇禽怪兽，穷诸灵异，皆自然运动。汉人焚香之器至博山炉而大备，后世代有制作。', poem: { text: '博山炉中沉香火，双烟一气凌紫霞。', title: '杨叛儿', author: '唐 · 李白', }, ritual: '民间上香以三炷为常，香火不断为敬；烟直上者有「诚心直达」之说。', authenticity: 'canon', }, { id: 'candle', category: 'vessel', name: '烛台', poeticTitle: '双烛长明', dynasty: '汉魏', origin: '战国以来烛具，汉魏成制', allusion: '古人以麻蒸、蜂蜡为烛，战国已有烛具，汉魏渐成台式。祠堂祭案设烛台一对，长明不灭，以照幽冥。', poem: { text: '蜡烛有心还惜别，替人垂泪到天明。', title: '赠别', author: '唐 · 杜牧', }, ritual: '祭案烛台成对，左右各一，取阴阳相配、昼夜长明之意。', authenticity: 'canon', }, { id: 'tea_cups', category: 'vessel', name: '茶盏', poeticTitle: '清茶三献', dynasty: '宋', origin: '宋 · 蔡襄《茶录》',

allusion: '宋人点茶尚黑釉盏，蔡襄《茶录》论盏色：「茶色白，宜黑盏」，建安兔毫盏遂为斗茶上品。祭案设茶，则取其清而不取其味。', poem: { text: '一碗喉吻润，两碗破孤闷。', title: '走笔谢孟谏议寄新茶', author: '唐 · 卢仝', }, ritual: '祭案常设清茶三盏，与酒果并陈，谓之「清酌庶羞」之余意。', authenticity: 'canon', }, { id: 'singing_bowl', category: 'vessel', name: '铜磬', poeticTitle: '一声清磬', dynasty: '唐', origin: '佛门呗器，历代沿袭', allusion: '磬本为石制乐器，入佛门则为铜钵、铜磬，为呗器之一，击之以集众、以收心。清音既起，万籁俱寂。', poem: { text: '万籁此都寂，但余钟磬音。', title: '题破山寺后禅院', author: '唐 · 常建', }, ritual: '击磬一响，以净心神，多在献香、献茶前后行之。', authenticity: 'canon', }, { id: 'vase', category: 'vessel', name: '供瓶', poeticTitle: '一花一器', dynasty: '明', origin: '明 · 袁宏道《瓶史》', allusion: '袁宏道《瓶史》专论插花之器与法，谓插花不可太繁、亦不可太瘦，多不过二种三种，高低疏密如画苑布置方妙。瓶花清供，至明而成一家之学。', poem: { text: '插花不可太繁，亦不可太瘦，多不过二种三种。', title: '瓶史 · 宜称', author: '明 · 袁宏道', }, ritual: '供瓶成对称于香炉两侧，花头朝内，取「束香与花以献」之礼。', authenticity: 'canon', related: ['peonies', 'lotus'], }, { id: 'melon_lobed_celadon', category: 'vessel', name: '瓜棱青瓷瓶', poeticTitle: '分瓣含翠',

dynasty: '宋', origin: '取法宋金瓜棱瓶（龙泉窑、耀州窑一系）', allusion: '瓜棱瓶以竖凹分瓣如瓜，宋金南北窑口皆有烧造，青釉厚处凝翠、薄处见胎，为宋人尚「类玉」之器。', note: '此为今器取法宋式，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'ribbed_fluted_white', category: 'vessel', name: '竖棱白釉瓶', poeticTitle: '甜白出筋', dynasty: '明', origin: '取法元代枢府釉模印出筋与明永乐甜白釉', allusion: '甜白釉明永乐时为极品，釉色莹润如凝脂；出筋之法，则于胎上起棱，釉薄处微见骨色，白中见骨，最耐看。', note: '此为今器取法明清白瓷，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'fluted_trapezoid_mini', category: 'vessel', name: '束腰小尊', poeticTitle: '觚哉小尊', dynasty: '商周', origin: '取法商周青铜觚之束腰外撇', allusion: '觚为商周饮酒之器，喇叭口、细腰、外撇足，孔子有「觚不觚，觚哉觚哉」之叹，惜其失制。后世花器多仿觚形，小者宜案头清供。', note: '此为今器取法青铜觚的基本轮廓，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'chiseled_stone_gray', category: 'vessel', name: '凿石纹瓶', poeticTitle: '凿石留痕', dynasty: '汉', origin: '取法汉代石雕与画像石的凿刻肌理', allusion: '汉画像石以减地平雕见长，凿痕成行如梳，拙而不野。今以凿刻肌理入陶，取其厚重肃穆，宜配松枝、白菊。', note: '此为今器取法汉石作肌理，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'faceted_gem_prism', category: 'vessel',

name: '多面琉璃器', poeticTitle: '棱光分色', dynasty: '隋唐', origin: '取法隋唐西亚玻璃器（法门寺地宫所出其一系）', allusion: '隋唐时西域玻璃器入贡，多面磨花、壁薄色清，为内库珍玩。今以多棱折面拟之，光过成彩。', note: '此为今器取法古代玻璃器折光之趣，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'gradient_sand_volcano', category: 'vessel', name: '窑变砂釉器', poeticTitle: '入窑一色', dynasty: '宋', origin: '取法宋钧窑窑变', allusion: '钧窑以铜红釉入窑，火之所在，色之所在，故有「入窑一色，出窑万彩」之说。釉色自上而下流淌，如砂丘、如熔岩。', note: '器型为当代设计，取法在釉色之变，非宋钧原器。', authenticity: 'canon', related: ['vase'], }, { id: 'pinched_clay_ruffle', category: 'vessel', name: '泥捏翻边陶瓶', poeticTitle: '泥痕翻卷', dynasty: '新石器', origin: '取法彩陶泥条盘筑与梳齿纹（马家窑一系）', allusion: '彩陶以泥条盘筑成形，指腹按压处留痕，口沿翻卷如荷叶，梳齿划纹成行。拙手之痕即为纹，最见初民之意。', note: '此为今器取法史前陶艺手法，非出土原物。', authenticity: 'canon', related: ['vase'], }, { id: 'twisted_helix_tower', category: 'vessel', name: '长颈旋纹瓶', poeticTitle: '修颈亭亭', dynasty: '宋元', origin: '取法宋元经瓶（今称梅瓶）之小口修颈', allusion: '小口、丰肩、修颈而下敛者，宋人谓之经瓶，为酒器；后世雅称梅瓶，以其口径小仅容梅枝故。今变其比例作修长塔式。', note: '此为今器取法宋元瓶式比例，非出土原物。', authenticity: 'canon', related: ['vase'], }, {

id: 'organic_twist_cream', category: 'vessel', name: '卷泥素瓶', poeticTitle: '素手卷泥', dynasty: '当代', origin: '当代手工陶艺', allusion: '当代陶艺借手捏拉坯，令器身微旋成势，素色无纹，重在手痕与釉面的先后对话。', note: '当代设计器型，无古代器式可考。', authenticity: 'modern', related: ['vase'], }, { id: 'organic_twist_rose', category: 'vessel', name: '卷泥胭瓶', poeticTitle: '胭色微醺', dynasty: '当代', origin: '当代手工陶艺', allusion: '同上器之制，施胭脂色釉，深浅随釉层厚薄而变，宜配浅色花材。', note: '当代设计器型，无古代器式可考。', authenticity: 'modern', related: ['vase'], }, { id: 'minimal_drop_swan', category: 'vessel', name: '水滴素瓶', poeticTitle: '一滴静水', dynasty: '当代', origin: '当代极简设计', allusion: '取水滴将落未落之势，颈细腹圆，不施纹饰，以形取胜，为当代极简一派。', note: '当代设计器型，无古代器式可考。', authenticity: 'modern', related: ['vase'], }, { id: 'pearlescent_turbine', category: 'vessel', name: '珠光涡轮瓶', poeticTitle: '珠光流转', dynasty: '当代', origin: '当代设计，取珠光釉面', allusion: '以珠光釉随器身旋起，光行如水涡，属当代陈设器，重光影而不重纹样。', note: '当代设计器型，无古代器式可考。', authenticity: 'modern', related: ['vase'],

},];
/* ===== 源文件：src/data/sceneItems.ts ===== */ /**

- @file sceneItems.ts
- @description 馆空间（堂轩）供案陈设的纯数据层：摆件工厂、初始陈设、审计展台与常量字典。

*

- 本模块零 UI 依赖（仅引用类型与一个纯函数工具），可独立单测：
- 输入确定的参数即得到确定的 SceneItem 集合，不含任何 React 状态与副作用。

*/ import { SceneItem, SceneItemId, LightingSettings, MemorialInfo, MemorialHallItem, FlowerType, FruitType, TableWoodStyle, AuditCategory, AuditGroup, VaseAuditGroup, FlowerAuditGroup, } from '../types'; import type { VaseStyle } from '../pixi'; import { resolveSystemTimeOfDay } from '../utils/timeOfDay';
/* —— 摆件构造器 —— */
export const createVaseItem = (id: string, flowerType: FlowerType | 'none', position: { x: number; y: number }, vaseStyle: VaseStyle = 'warm_white_plain', name: string = '花瓶', scale: number = 0.95): SceneItem => ({ id, type: 'vase', name, flowerType: flowerType as FlowerType, vaseStyle, position, scale, });
export const createBouquetItem = (flowerType: FlowerType, position: { x: number; y: number },

options?: { id?: string; name?: string }): SceneItem => ({ id: options?.id || 'bouquet', type: 'flower', name: options?.name || 敬献 · ${flowerType}, flowerType, position, scale: 0.95, });

/* —— 名目字典 —— */

/** 点击花冠时循环切换的花材顺序 */ export const FLOWER_CYCLE_ORDER: FlowerType[] = ['peonies', 'lilies', 'lotus', 'chrysanthemum', 'roses', 'plum_blossoms', 'sunflowers', 'carnations', 'calla_lilies', 'tulips', 'narcissus', 'lily_of_valley', 'orchids', 'hydrangeas', 'daisies',];

/** 点击果盘时循环切换的果品顺序 */ export const FRUIT_CYCLE_ORDER: FruitType[] = ['apples', 'oranges', 'peaches', 'persimmons', 'citrons', 'grapes', 'pomelos', 'longans', 'bananas', 'kumquats', 'dragonfruits', 'cucumbers', 'watermelons', 'coconuts', 'plums', 'mangosteens', 'pineapples',

'pears',];

/** 点击瓶身时循环切换的器形顺序 */ export const VASE_STYLE_CYCLE_ORDER: VaseStyle[] = ['warm_white_plain', 'painted_iris', 'painted_peony', 'painted_plum', 'qinghua_lotus', 'royal_blue_gold', 'mutton_jade_ears', 'celadon_ru',];

export const FLOWER_DISPLAY_NAMES: Record<FlowerType, string> = { peonies: '富贵牡丹', lilies: '纯白百合', lotus: '出尘净莲', chrysanthemum: '岁寒金菊', roses: '长情玫瑰', plum_blossoms: '傲霜冷香白梅', sunflowers: '金色向阳', carnations: '慈孝康乃馨', calla_lilies: '圣洁马蹄莲', tulips: '典雅郁金香', narcissus: '凌波水仙', lily_of_valley: '如意铃兰', orchids: '幽谷素兰', hydrangeas: '团圆绣球', daisies: '清纯雏菊', };

export const FRUIT_DISPLAY_NAMES: Record<FruitType, string> = { apples: '平安红富士苹果', oranges: '吉祥金蜜柑', peaches: '福寿仙桃', persimmons: '如意熟丹柿', citrons: '禅意佛手柑', grapes: '阳光玫瑰青提', pomelos: '黄金蜜柚', longans: '富贵龙眼', bananas: '招财金蕉', kumquats: '翡翠青桔', dragonfruits: '红运火龙果', cucumbers: '翠玉黄瓜', watermelons: '盛夏西瓜', coconuts: '南国椰子', plums: '金玉蜂糖李', mangosteens: '果后山竹',

pineapples: '祥瑞凤梨', pears: '金秋秋月梨', };

/** 供案 4 款原木材质选项（台面与腿足构架共用） */ export const TABLE_WOOD_OPTIONS = [{ id: 'walnut', name: '暖栗胡桃', color: '#452618', border: '#7c3f25' }, { id: 'huanghuali', name: '明式黄花梨', color: '#7a421b', border: '#b4602a' }, { id: 'cherry', name: '典雅樱桃', color: '#6a2f20', border: '#9a4731' }, { id: 'white_oak', name: '素雅白橡', color: '#8c6e54', border: '#b8997a' },] as const;

export type TableWoodOptionId = (typeof TABLE_WOOD_OPTIONS)[number]['id'];

/* —— 陈设

---

*/

/** 供案常规陈设（五供齐备：香、花、灯、水、果） */ export const buildStandardItems = (): Record<SceneItemId, SceneItem> => ({ vase_left: createVaseItem('vase_left', 'peonies', { x: -164, y: -4 }, 'warm_white_plain', '富贵牡丹花瓶（左）'), vase_right: createVaseItem('vase_right', 'lilies', { x: 164, y: -4 }, 'painted_iris', '纯洁百合花瓶（右）'), candle_left: { id: 'candle_left', type: 'candle', name: '长明红烛（左）', position: { x: -74, y: 3 }, isLit: true, }, candle_right: { id: 'candle_right', type: 'candle', name: '长明红烛（右）', position: { x: 74, y: 3 }, isLit: true, }, censer: { id: 'censer', type: 'censer', name: '青瓷清香香炉（中）', position: { x: 0, y: 6 }, isLit: true, }, fruit_left: { id: 'fruit_left', type: 'fruit', name: '平安红富士苹果盘（左）', fruitType: 'apples', position: { x: -118, y: 16 }, }, fruit_right: { id: 'fruit_right',

type: 'fruit', name: '吉祥砂糖橘盘 (右)', fruitType: 'oranges', position: { x: 118, y: 16 }, }, tea_cups: { id: 'tea_cups', type: 'tea_cups', name: '虔诚敬茶三盏', position: { x: 0, y: 22 }, }, singing_bowl: { id: 'singing_bowl', type: 'singing_bowl', name: '静心铜磬颂钵 (左前)', position: { x: -62, y: 32 }, }, });

export type { AuditCategory, VaseAuditGroup, FlowerAuditGroup, AuditGroup };

/** 1. 雅瓷花器形制专项审计 */ export const buildAuditVaseItems = (group: VaseAuditGroup = 'new_unique', withFlowers: boolean = true): Record<SceneItemId, SceneItem> => { if (group === 'new_unique') { // 4 款全新异形与手作/旋涡/流沙微观质感花器 (最新参考图提炼打造) return { vase_audit_1: createVaseItem('vase_audit_1', withFlowers ? 'peonies' : 'none', { x: -140, y: 0 }, 'pinched_clay_ruffle', '1. 手作泥捏翻边陶瓶 (梳齿拉坯刮痕粗陶)', 0.96), vase_audit_2: createVaseItem('vase_audit_2', withFlowers ? 'lilies' : 'none', { x: -46, y: -4 }, 'pearlescent_turbine', '2. 银曜珠光螺旋涡轮瓶 (8道3D旋转刀锋棱)', 0.96), vase_audit_3: createVaseItem('vase_audit_3', withFlowers ? 'lotus' : 'none', { x: 46, y: -4 }, 'twisted_helix_tower',

'3. 极简莫兰迪旋转扭扭塔 (90° 沙漏扭转腰)', 0.96), vase_audit_4: createVaseItem('vase_audit_4', withFlowers ? 'roses' : 'none', { x: 140, y: 0 }, 'gradient_sand_volcano', '4. 天青流沙渐变圆锥瓶 (砂岩颗粒极简锥台)', 0.96), }; }

// designer_twist: 8 款现代几何折痕与器形 return { vase_audit_1: createVaseItem('vase_audit_1', withFlowers ? 'peonies' : 'none', { x: -189, y: 4 }, 'ribbed_fluted_white', '1. 极细百褶密棱瓶', 0.82), vase_audit_2: createVaseItem('vase_audit_2', withFlowers ? 'lilies' : 'none', { x: -135, y: 0 }, 'organic_twist_cream', '2. 旋风波浪奶油瓶', 0.82), vase_audit_3: createVaseItem('vase_audit_3', withFlowers ? 'peonies' : 'none', { x: -81, y: -4 }, 'organic_twist_rose', '3. 落樱粉旋风波纹瓶', 0.82), vase_audit_4: createVaseItem('vase_audit_4', withFlowers ? 'roses' : 'none', { x: -27, y: -6 }, 'chiseled_stone_gray', '4. 侘寂砂岩横纹瓶', 0.82), vase_audit_5: createVaseItem('vase_audit_5', withFlowers ? 'lotus' : 'none', { x: 27, y: -6 }, 'fluted_trapezoid_mini', '5. 圆台百褶梯形瓶', 0.82), vase_audit_6: createVaseItem('vase_audit_6', withFlowers ? 'chrysanthemum' : 'none', { x: 81, y: -4 }, 'faceted_gem_prism', '6. 钻石晶体切角瓶', 0.82), vase_audit_7: createVaseItem('vase_audit_7', withFlowers ? 'lilies' : 'none', { x: 135, y: 0 }, 'melon_lobed_celadon', '7. 海棠瓜棱秀颈瓶', 0.82), vase_audit_8: createVaseItem('vase_audit_8', withFlowers ? 'lotus' : 'none', { x: 189, y: 4 }, 'minimal_drop_swan', '8. 极简水滴天鹅瓶', 0.82), }; };

/** 2. 🌸 名花形神与枯荣生命周期专项审计 */ export const buildAuditFlowerItems = (group: FlowerAuditGroup = 'flowers_tier1_all', decayHealth: number = 1.0): Record<SceneItemId, SceneItem> => { const lifecycle = { stage: (decayHealth < 0.2 ? 'withered' : decayHealth < 0.7 ? 'decaying' : 'blooming') as any, age: (1 - decayHealth) * 200, duration: 240, decayRate: 1.0, health: decayHealth, autoDespawn: false, };

if (group === 'flowers_tier1_a') { // 5 款第一梯队名花 (前5) return { vase_audit_flower_1: { ...createVaseItem('vase_audit_flower_1', 'plum_blossoms', { x: -144, y: 0 }, 'ribbed_fluted_white',

'1. 傲霜冷香白梅 · 枯瘦挺拔细折枝', 0.96), lifecycle, }, vase_audit_flower_2: { ...createVaseItem('vase_audit_flower_2', 'sunflowers', { x: -72, y: -4 }, 'chiseled_stone_gray', '2. 金色灿烂向日 · 葵盘蜂巢蕊', 0.96), lifecycle, }, vase_audit_flower_3: { ...createVaseItem('vase_audit_flower_3', 'carnations', { x: 0, y: -6 }, 'organic_twist_rose', '3. 慈孝粉康乃馨 · 锯齿折叠瓣', 0.96), lifecycle, }, vase_audit_flower_4: { ...createVaseItem('vase_audit_flower_4', 'calla_lilies', { x: 72, y: -4 }, 'minimal_drop_swan', '4. 素雅圣洁马蹄莲 · 纯白佛焰苞', 0.96), lifecycle, }, vase_audit_flower_5: { ...createVaseItem('vase_audit_flower_5', 'tulips', { x: 144, y: 0 }, 'pearlescent_turbine', '5. 优雅高脚红郁金香 · 艳红高杯', 0.96), lifecycle, }, };

```
}
if (group === 'flowers_tier1_b') { // 5 款第一梯队名花 (后5) return { vase_audit_flower_6: { ...createVaseItem( 'vase_audit_flower_6', 'narcissus', { x: -144, y: 0 }, 'melon_lobed_celadon', '6. 凌波仙子水仙 · 金盏银台', 0.96 ), lifecycle, }, vase_audit_flower_7: { ...createVaseItem( 'vase_audit_flower_7', 'lily_of_valley', { x: -72, y: -4 }, 'fluted_trapezoid_mini', '7. 幽谷如意铃兰 · 拱枝乳白银铃', 0.96 ), lifecycle, }, vase_audit_flower_8: { ...createVaseItem( 'vase_audit_flower_8', 'orchids', { x: 0, y: -6 }, 'twisted_helix_tower', '8. 空谷幽兰蝴蝶兰 · 展翅粉紫唇', 0.96 ), lifecycle, }, vase_audit_flower_9: { ...createVaseItem( 'vase_audit_flower_9', 'hydrangeas', { x: 72, y: -4 }, 'pinched_clay_ruffle', '9. 团圆锦簇紫绣球 · 繁英聚球', 0.96 ), lifecycle, }, vase_audit_flower_10: {
```

暖念纪念场景模拟软件 V1.0 · 源程序（后30页，共60页） 说明：每页不少于50行；页眉为软件名称与版本号，右下角为页码。

{track.icon} {track.name} {track.poeticName} </button>); })}

{/* 背景音乐音量控制条 */} 乐曲音量 <input type="range" min="0" max="1" step="0.01" value={bgmState.volume} onChange={(e) => sound.setBgmVolume(parseFloat(e.target.value))} className="flex-1 accent-amber-500 h-1.5 bg-stone-800 rounded-lg cursor-pointer" /> {Math.round(bgmState.volume * 100)}%

{/* —— 9. 摆件交互音效开关 —— */}   {soundEnabled ? (<Volume2 size={16} className="text-amber-400 shrink-0" />) : (<VolumeX size={16} className="text-stone-500 shrink-0" />)} 祭祀与供奉触碰音效（敲磬/点烛/置杯） <button type="button" onClick={onToggleSound} className={`px-3 py-1 rounded-lg text-xs font-medium transition-colors cursor-pointer border shrink-0 whitespace-nowrap ${ soundEnabled ? 'bg-amber-600 border-amber-400/60 text-amber-950 font-bold' : 'bg-stone-800 border-stone-700 text-stone-400' }`}

{soundEnabled ? '开' : '关'} </button>

{/* —— 10. 专项展台审计开关与深度配置（支持展开折叠） —— */}

{onToggleAuditMode && ({/* 头部开关与展开折叠控制 */} <div className="flex items-center justify-between cursor-pointer select-none gap-2.5" onClick={() => { sound.playWoodPlacement(); setIsAuditExpanded((prev) => !prev); }}

 <div className={w-8 h-8 rounded-lg flex items-center justify-center shrink-0 border transition-all ${ isAuditMode ? 'bg-amber-500/20 border-amber-400 text-amber-300 shadow-[0_0_12px_rgba(245,158,11,0.3)]' : 'bg-stone-800 border-stone-700 text-stone-500' }}> <Eye size={16} />   专项展台审计（名花与花器全景品鉴） {isAuditExpanded ? '收起配置' : '展开配置'} {isAuditExpanded ? <ChevronUp size={11} /> : <ChevronDown size={11} />}   {isAuditMode ? '展台已激活 · 点击可折叠或微调审计大类、分组与枯荣度' : '开启后在案台呈现可自由拖拽的全景名花与名瓷品鉴展台'}

<div className="flex items-center gap-2 shrink-0 ml-1" onClick={(e) => e.stopPropagation()}> {/* 开关按钮 */} <button type="button" onClick={() => { sound.playWoodPlacement(); onToggleAuditMode(); if (!isAuditMode) { setIsAuditExpanded(true); } }} className={px-3 py-1.5 rounded-lg text-xs font-medium transition-colors cursor-pointer border shrink-0 whitespace-nowrap ${ isAuditMode ? 'bg-gradient-to-r from-rose-600 to-amber-600 border-amber-400/60 text-stone-950 font-bold shadow-xs' : 'bg-stone-800 border-stone-700 text-stone-400 hover:text-stone-200' }}

{isAuditMode ? '已开启' : '开启'}

</button>

{/* 展开后的详细配置面板 */} {isAuditExpanded && ({/* 1. 品鉴大类切换 (名花 vs 花器 vs 供案) */} <label className="text-[11px] font-medium text-amber-200/90 flex items-center justify-between font-serif"> 品鉴大类 {auditCategory === 'flowers' ? '当前：15款名卉形神品鉴' : '当前：12款雅瓷花器形制'} </label> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditCategory?.('flowers'); }} className={`py-1.5 px-2.5 rounded-xl border text-xs font-medium flex items-center justify-center gap-1.5 cursor-pointer transition-all ${ auditCategory === 'flowers' ? 'bg-amber-500 text-stone-950 font-bold border-amber-400 shadow-sm' : 'bg-stone-950/70 border-stone-800 text-stone-300 hover:bg-stone-800/80 hover:text-stone-100' }`}

🌸 名花形神专项 (15款) </button>

<button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditCategory?.('vases'); }} className={`py-1.5 px-2.5 rounded-xl border text-xs font-medium flex items-center justify-center gap-1.5 cursor-pointer transition-all ${ auditCategory === 'vases' ? 'bg-amber-500 text-stone-950 font-bold border-amber-400 shadow-sm' : 'bg-stone-950/70 border-stone-800 text-stone-300 hover:bg-stone-800/80 hover:text-stone-100' }`}

🏺 花器形制专项 (12款) </button>

<button type="button"

onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditCategory?.('table'); }} className={`py-1.5 px-2.5 rounded-xl border text-xs font-medium flex items-center justify-center gap-1.5 cursor-pointer transition-all ${ auditCategory === 'table' ? 'bg-amber-500 text-stone-950 font-bold border-amber-400 shadow-sm' : 'bg-stone-950/70 border-stone-800 text-stone-300 hover:bg-stone-800/80 hover:text-stone-100' }`}

🪵 供案原木材质 (4款) </button>

{/* 2. 展台阵列分组 */} <label className="text-[11px] font-medium text-amber-200/90 font-serif"> {auditCategory === 'flowers' ? '名花陈列分组' : '雅瓷花器分组'} </label> {auditCategory === 'flowers' ? (<button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('flowers_tier1_all'); }} className={`py-1.5 px-2 rounded-lg border text-[11px] text-center cursor-pointer transition-all ${ auditGroup === 'flowers_tier1_all' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }`}

🌟 10花齐放同台 </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('flowers_tier1_a'); }} className={`py-1.5 px-2 rounded-lg border text-[11px] text-center cursor-pointer transition-all ${ auditGroup === 'flowers_tier1_a' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }`}

🌺 梯队A (前5)

</button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('flowers_tier1_b'); }} className={`py-1.5 px-2 rounded-lg border text-[11px] text-center cursor-pointer transition-all ${ auditGroup === 'flowers_tier1_b' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }`}

🪷 梯队B (后5) </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('flowers_classic'); }} className={`py-1.5 px-2 rounded-lg border text-[11px] text-center cursor-pointer transition-all ${ auditGroup === 'flowers_classic' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }`}

💐 经典5供名花 </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('flowers_bouquets'); }} className={`py-1.5 px-2 rounded-lg border text-[11px] text-center cursor-pointer transition-all col-span-2 sm:col-span-1 ${ auditGroup === 'flowers_bouquets' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }`}

💒 案前敬献花束 </button>) : (<button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.();

onSwitchAuditGroup?.('new_unique'); }} className={`py-1.5 px-2.5 rounded-lg border text-xs text-center cursor-pointer transition-all ${ auditGroup === 'new_unique' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }}

✨ 4款异形肌理瓶 </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSwitchAuditGroup?.('designer_twist'); }} className={`py-1.5 px-2.5 rounded-lg border text-xs text-center cursor-pointer transition-all ${ auditGroup === 'designer_twist' ? 'bg-amber-500/25 border-amber-400 text-amber-200 font-bold ring-1 ring-amber-400/40' : 'bg-stone-950/50 border-stone-800 text-stone-400 hover:text-stone-200 hover:bg-stone-800/50' }}

🎨 8款现代折痕瓶 </button>)}

{/* 3. 详细调节项：名花生命枯荣度 & 裸检模式 */} {/* 名花生命枯荣调节 */} {auditCategory === 'flowers' && (<label className="text-[10px] text-amber-200/80 font-medium flex items-center justify-between"> 枯荣生命阶段 {Math.round((flowerDecayHealth ?? 1) * 100)}% </label> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSelectFlowerDecayHealth?.(1.0); }} className={`py-1 px-1 rounded text-[10px] text-center cursor-pointer transition-all border ${ (flowerDecayHealth ?? 1) >= 0.9 ? 'bg-rose-950/60 border-rose-400 text-rose-200 font-bold' : 'bg-stone-900 border-stone-800 text-stone-400 hover:text-stone-200'

}`}

🌸 满盛 </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSelectFlowerDecayHealth?.(0.55); }} className={py-1 px-1 rounded text-[10px] text-center cursor-pointer transition-all border ${ (flowerDecayHealth ?? 1) > 0.3 && (flowerDecayHealth ?? 1) < 0.9 ? 'bg-amber-950/60 border-amber-400 text-amber-200 font-bold' : 'bg-stone-900 border-stone-800 text-stone-400 hover:text-stone-200' }}

🍂 微枯 </button> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onSelectFlowerDecayHealth?.(0.15); }} className={py-1 px-1 rounded text-[10px] text-center cursor-pointer transition-all border ${ (flowerDecayHealth ?? 1) <= 0.3 ? 'bg-stone-800 border-stone-600 text-stone-300 font-bold' : 'bg-stone-900 border-stone-800 text-stone-400 hover:text-stone-200' }}

🥀 岁暮 </button>   )}

{/* 插花景深 vs 裸检切换 */} <div className={space-y-1.5 p-2 bg-stone-950/60 rounded-xl border border-stone-800 ${ auditCategory === 'flowers' ? '' : 'sm:col-span-2' }}> <label className="text-[10px] text-amber-200/80 font-medium"> 瓶口景深与微观肌理 </label> <button type="button" onClick={() => { if (!isAuditMode) onToggleAuditMode?.(); onToggleAuditFlowers?.(); }} className="w-full py-1.5 px-2 rounded-lg bg-amber-950/30 hover:bg-amber-900/40 text-amber-200 border border-amber-500/40 cursor-pointer text-xs font-medium flex items-center justify-center gap-1.5 transition-all"

{auditWithFlowers ? '🌸 切换为净瓶裸检（看肌理）' : '🌸 恢复插花入瓶（看咬合）'} </button>

{/* 快捷操作条：触发落英 / 恢复常规五供 */} {auditCategory === 'flowers' && onTriggerAuditPetals && (<button type="button" onClick={() => { onTriggerAuditPetals(); }} className="text-amber-300 hover:text-amber-200 flex items-center gap-1 cursor-pointer transition-colors"

🍃 触发漫天落瓣动效 </button>)} {isAuditMode && (<button type="button" onClick={() => { onToggleAuditMode?.(); }} className="text-stone-400 hover:text-amber-300 flex items-center gap-1 cursor-pointer ml-auto transition-colors"

<RotateCcw size={12} /> 一键退出审计 · 恢复常规五供 </button>)}   )})}

{/* 协议与立意公约说明入口 */} <button type="button" onClick={() => { close(); onOpenAgreement?.(); }} className="flex items-center gap-1.5 text-amber-300/80 hover:text-amber-200 cursor-pointer transition-colors"

<ShieldCheck size={14} className="text-amber-400" /> 《暖念服务协议、立意与收费声明》 </button>

v1.0.0 官方公约

{/* Pinned Bottom Confirm Bar */}   全部调节已实时生效 · 素心寄思 <button type="button" onClick={close} className="px-6 py-2 rounded-xl bg-gradient-to-r from-amber-600 to-amber-700 hover:from-amber-500 hover:to-amber-600 active:scale-95 text-stone-950 font-bold text-xs shadow-lg border border-amber-400/60 transition-all cursor-pointer flex items-center gap-1.5"

<Check size={14} /> 完成确认 </button>   ); };

export const LightingPanel = AmbiancePanel;

/* ===== 源文件：src/ui/space/setting/index.ts ===== */ /**

- @file index.ts
- @description 氛围与场景设置特性统一导出入口。

*/ export * from './AmbiancePanel'; export * from './openAmbianceDrawer';

/* ===== 源文件：src/ui/space/setting/openAmbianceDrawer.ts ===== */ /**

- @file openAmbianceDrawer.ts
- @description 氛围与场景设置抽屉的打开入口（右侧抽屉形态）。

*/ import type { ComponentProps } from 'react'; import { openDialog } from '../../common/dialog'; import { AmbiancePanel } from './AmbiancePanel';

export const openAmbianceDrawer = (props: ComponentProps<typeof AmbiancePanel>) => openDialog(AmbiancePanel, props, { variant: 'drawer', width: '30rem', closeOnBackdrop: true });

/* ===== 源文件：src/utils/audio.ts ===== */ /**

- @file audio.ts
- @description Web Audio API 纯原生算法合成音效与禅意背景音乐引擎（铜磬、木鱼、甘露水滴、微烛与4大禅音背景音乐）。

*/ import { BgmTrackId, BgmTrackInfo } from '../types';

export const BGM_TRACKS: BgmTrackInfo[] = [{ id: 'singing_bowl', name: '空灵钵韵', poeticName: '528Hz 祈福疗愈', instrument: '纯铜颂钵 · 泛音磬钟', desc: '528Hz 祈愿频率，深层共鸣钵体余音悠长，令人心神宁静澄澈。', icon: '🥣', }, { id: 'bamboo_wind', name: '幽篁清风', poeticName: '修竹甘露 · 禅寺木鱼', instrument: '竹林微风 · 清泉水滴 · 静心木鱼', desc: '微风轻拂青竹，偶有晨露滴落与沉稳木鱼声，如临山间精舍。', icon: '🎋', }, { id: 'temple_bell', name: '晨钟远山', poeticName: '梵音古刹 · 钟声悠扬', instrument: '远山大钟 · 沉香梵唱', desc: '庄严肃穆的名山古刹晨钟，低回深远，寄托无限怀思。', icon: '🔔', }, { id: 'pentatonic_zither', name: '五音琴韵', poeticName: '高山流水 · 宫商角徵羽', instrument: '七弦古琴 · 纯净泛音', desc: '中国传统五声音阶断续幽微，如山涧流泉、春蚕丝雨。', icon: '🎻', },];

class SoundEngine { private ctx: AudioContext | null = null; public enabled: boolean = true;

// —— 背景音乐（BGM）引擎控制 —— public bgmPlaying: boolean = false; public currentTrackId: BgmTrackId = 'singing_bowl'; public bgmVolume: number = 0.55; private bgmGain: GainNode | null = null; private bgmIntervalId: number | null = null; private bgmNodes: (AudioNode | OscillatorNode)[] = []; private bgmListeners: Array<(state: { isPlaying: boolean; trackId: BgmTrackId; volume: number }) => void> = [];

private init() {

```
if (!this.ctx && typeof window !== 'undefined') { const AudioCtx = window.AudioContext || (window as unknown as { webkitAudioContext: typeof AudioContext }).webkitAudioContext; this.ctx = new AudioCtx(); } if (this.ctx && this.ctx.state === 'suspended') { this.ctx.resume(); } }

// 订阅 BGM 播放状态更新 public subscribeBgm(listener: (state: { isPlaying: boolean; trackId: BgmTrackId; volume: number }) => void) { this.bgmListeners.push(listener); listener({ isPlaying: this.bgmPlaying, trackId: this.currentTrackId, volume: this.bgmVolume }); return () => { this.bgmListeners = this.bgmListeners.filter((l) => l !== listener); }; }

private notifyBgmListeners() { this.bgmListeners.forEach((fn) => fn({ isPlaying: this.bgmPlaying, trackId: this.currentTrackId, volume: this.bgmVolume }) ); }

// 设置 BGM 整体音量 public setBgmVolume(val: number) { this.bgmVolume = Math.max(0, Math.min(1, val)); if (this.bgmGain && this.ctx) { this.bgmGain.gain.setValueAtTime(this.bgmVolume, this.ctx.currentTime); } this.notifyBgmListeners(); }

// 切换播放/暂停 public toggleBgm() { if (this.bgmPlaying) { this.stopBgm(); } else { this.playBgm(this.currentTrackId); } }

// 切换到下一首曲目 public nextBgmTrack() { const idx = BGM_TRACKS.findIndex((t) => t.id === this.currentTrackId); const nextIdx = (idx + 1) % BGM_TRACKS.length; this.playBgm(BGM_TRACKS[nextIdx].id); }

// 播放指定曲目
```

```
public playBgm(trackId: BgmTrackId = this.currentTrackId) { this.init(); if (!this.ctx) return;

this.stopBgm(false); this.currentTrackId = trackId; this.bgmPlaying = true;

// 建立总主控音量节点 this.bgmGain = this.ctx.createGain(); this.bgmGain.gain.setValueAtTime(0, this.ctx.currentTime); this.bgmGain.gain.linearRampToValueAtTime(this.bgmVolume, this.ctx.currentTime + 1.2); this.bgmGain.connect(this.ctx.destination);

// 根据选定的禅音曲目启动原生合成声景 switch (trackId) { case 'singing_bowl': this.startSingingBowlAmbient(); break; case 'bamboo_wind': this.startBambooWindAmbient(); break; case 'temple_bell': this.startTempleBellAmbient(); break; case 'pentatonic_zither': this.startPentatonicZitherAmbient(); break; default: this.startSingingBowlAmbient(); }

this.notifyBgmListeners(); }

// 停止播放 BGM public stopBgm(notify: boolean = true) { if (this.bgmIntervalId !== null) { window.clearInterval(this.bgmIntervalId); this.bgmIntervalId = null; }

if (this.bgmGain && this.ctx) { try { this.bgmGain.gain.linearRampToValueAtTime(0.0001, this.ctx.currentTime + 0.5); } catch { // ignore } }
```

```
// 延时关闭并断开所有节点 setTimeout(() => { this.bgmNodes.forEach((node) => { try { if ('stop' in node && typeof (node as OscillatorNode).stop === 'function') { (node as OscillatorNode).stop(); } node.disconnect(); } catch { // ignore } }); this.bgmNodes = []; }, 550);

this.bgmPlaying = false; if (notify) { this.notifyBgmListeners(); } }

// —— 1. 《空灵钵韵 · 528Hz》环境合成器 —— private startSingingBowlAmbient() { if (!this.ctx || !this.bgmGain) return; const now = this.ctx.currentTime;

// 528Hz 治愈温润持续声底 (Drone) const droneFreq = 132; // 528Hz 的次倍频基底 const osc1 = this.ctx.createOscillator(); const osc2 = this.ctx.createOscillator(); const gainDrone = this.ctx.createGain();

osc1.type = 'sine'; osc1.frequency.setValueAtTime(droneFreq, now); osc2.type = 'sine'; osc2.frequency.setValueAtTime(droneFreq * 1.5, now); // 纯五度泛音

// 慢速呼吸 LFO 调制 const lfo = this.ctx.createOscillator(); const lfoGain = this.ctx.createGain(); lfo.frequency.setValueAtTime(0.08, now); // 约 12 秒一个起伏呼吸周期 lfoGain.gain.setValueAtTime(0.06, now); lfo.connect(lfoGain);

gainDrone.gain.setValueAtTime(0.12, now); lfoGain.connect(gainDrone.gain);

osc1.connect(gainDrone); osc2.connect(gainDrone); gainDrone.connect(this.bgmGain);
```

```
osc1.start(now); osc2.start(now); lfo.start(now);

this.bgmNodes.push(osc1, osc2, lfo, gainDrone, lfoGain);

// 周期性轻磬敲击（每隔 8~10 秒发出轻柔的空灵余音） const triggerBowl = () => { if (!this.ctx || !this.bgmPlaying || !this.bgmGain) return; const t = this.ctx.currentTime; const bowlFreq = 528; const bOsc = this.ctx.createOscillator(); const bGain = this.ctx.createGain(); bOsc.type = 'sine'; bOsc.frequency.setValueAtTime(bowlFreq, t);

bGain.gain.setValueAtTime(0, t); bGain.gain.linearRampToValueAtTime(0.2, t + 0.05); bGain.gain.exponentialRampToValueAtTime(0.0001, t + 7.5);

bOsc.connect(bGain); bGain.connect(this.bgmGain); bOsc.start(t); bOsc.stop(t + 7.5); };

triggerBowl(); this.bgmIntervalId = window.setInterval(triggerBowl, 8500); }

// —— 2. 《幽篁清风 · 甘露竹影》环境合成器 —— private startBambooWindAmbient() { if (!this.ctx || !this.bgmGain) return;

// 竹林微风沙沙背景 (Filtered Pink/White Noise) const bufferSize = this.ctx.sampleRate * 2; const noiseBuffer = this.ctx.createBuffer(1, bufferSize, this.ctx.sampleRate); const output = noiseBuffer.getChannelData(0); let b0 = 0, b1 = 0, b2 = 0; for (let i = 0; i < bufferSize; i++) { const white = Math.random() * 2 - 1; b0 = 0.99886 * b0 + white * 0.0555179; b1 = 0.99332 * b1 + white * 0.0750759; b2 = 0.96900 * b2 + white * 0.1538520; output[i] = (b0 + b1 + b2) * 0.12; }

const noiseSource = this.ctx.createBufferSource(); noiseSource.buffer = noiseBuffer;
```

```
noiseSource.loop = true;
const windFilter = this.ctx.createBiquadFilter(); windFilter.type = 'lowpass'; windFilter.frequency.setValueAtTime(450, this.ctx.currentTime);
// 风声强弱呼吸调制 const windLfo = this.ctx.createOscillator(); windLfo.frequency.setValueAtTime(0.12, this.ctx.currentTime); const windLfoGain = this.ctx.createGain(); windLfoGain.gain.setValueAtTime(250, this.ctx.currentTime); windLfo.connect(windLfoGain); windLfoGain.connect(windFilter.frequency);
const windGain = this.ctx.createGain(); windGain.gain.setValueAtTime(0.15, this.ctx.currentTime);
noiseSource.connect(windFilter); windFilter.connect(windGain); windGain.connect(this.bgmGain);
noiseSource.start(); windLfo.start();
this.bgmNodes.push(noiseSource, windFilter, windGain, windLfo, windLfoGain);
// 随机清脆水滴声 (Dew Drip on Lotus / Pond) const triggerDrip = () => { if (!this.ctx || !this.bgmPlaying || !this.bgmGain) return; const t = this.ctx.currentTime; const dropFreq = 1600 + Math.random() * 800; const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'sine'; osc.frequency.setValueAtTime(dropFreq, t); osc.frequency.exponentialRampToValueAtTime(dropFreq * 1.8, t + 0.04);
gain.gain.setValueAtTime(0.12, t); gain.gain.exponentialRampToValueAtTime(0.001, t + 0.18);
osc.connect(gain); gain.connect(this.bgmGain); osc.start(t); osc.stop(t + 0.18); };
// 偶尔一声悠扬轻木鱼 const triggerWoodChime = () => { if (!this.ctx || !this.bgmPlaying || !this.bgmGain) return; const t = this.ctx.currentTime;
```

```
const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'triangle'; osc.frequency.setValueAtTime(220, t); osc.frequency.exponentialRampToValueAtTime(80, t + 0.08); gain.gain.setValueAtTime(0.1, t); gain.gain.exponentialRampToValueAtTime(0.001, t + 0.1); osc.connect(gain); gain.connect(this.bgmGain); osc.start(t); osc.stop(t + 0.1); };

this.bgmIntervalId = window.setInterval(() => { if (Math.random() > 0.4) triggerDrip(); if (Math.random() > 0.7) triggerWoodChime(); }, 2800); }

// —— 3. 《晨钟暮鼓 · 远山梵音》环境合成器 —— private startTempleBellAmbient() { if (!this.ctx || !this.bgmGain) return; const now = this.ctx.currentTime;

// 低频深邃大殿共振底噪 const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'sine'; osc.frequency.setValueAtTime(72, now); // 浑厚低音 gain.gain.setValueAtTime(0.08, now); osc.connect(gain); gain.connect(this.bgmGain); osc.start(now); this.bgmNodes.push(osc, gain);

// 周期性宏大远山古刹钟声 (Temple Bell Strike) const triggerBell = () => { if (!this.ctx || !this.bgmPlaying || !this.bgmGain) return; const t = this.ctx.currentTime; const baseFreq = 216; // 庄严钟声

// 基频 + 3 个金属泛音 [baseFreq, baseFreq * 1.62, baseFreq * 2.38, baseFreq * 3.4].forEach((freq, i) => { if (!this.ctx || !this.bgmGain) return; const o = this.ctx.createOscillator(); const g = this.ctx.createGain(); o.type = 'sine'; o.frequency.setValueAtTime(freq, t); g.gain.setValueAtTime(0, t); g.gain.linearRampToValueAtTime(0.18 / (i + 1), t + 0.04);
```

```
g.gain.exponentialRampToValueAtTime(0.0001, t + 9.5); o.connect(g); g.connect(this.bgmGain); o.start(t); o.stop(t + 9.5); }); };

triggerBell(); this.bgmIntervalId = window.setInterval(triggerBell, 10500); }

// —— 4.《五音琴韵 · 高山流水》中国古典五声音阶演算法 —— private startPentatonicZitherAmbient() { if (!this.ctx || !this.bgmGain) return;

// 中国传统五声音阶（宫 C4, 商 D4, 角 E4, 徵 G4, 羽 A4 及高八度 C5, D5, E5） const pentatonicScale = [261.63, 293.66, 329.63, 392.0, 440.0, 523.25, 587.33, 659.25];

const playZitherNote = (freq: number) => { if (!this.ctx || !this.bgmPlaying || !this.bgmGain) return; const t = this.ctx.currentTime; const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'triangle'; // 类似琴弦拨动明澈微泛音 osc.frequency.setValueAtTime(freq, t);

// 快速触弦与绵长泛音衰减 gain.gain.setValueAtTime(0, t); gain.gain.linearRampToValueAtTime(0.16, t + 0.02); gain.gain.exponentialRampToValueAtTime(0.0001, t + 3.8);

osc.connect(gain); gain.connect(this.bgmGain); osc.start(t); osc.stop(t + 3.8); };

let step = 0; const melodyPattern = [0, 2, 4, 3, 5, 2, 0, 4, 3, 1, 2, 4];

this.bgmIntervalId = window.setInterval(() => { const noteIdx = melodyPattern[step % melodyPattern.length]; const freq = pentatonicScale[noteIdx]; playZitherNote(freq); // 偶有双手和弦泛音 if (step % 3 === 0) { setTimeout(() => playZitherNote(freq * 1.5), 180); } step++;
```

}, 2200); }

// —— 交互离散音效 ——

// 轻音铜磬/钟磬 playBell(freq: number = 432, duration: number = 3.5) { if (!this.enabled) return; try { this.init(); if (!this.ctx) return;

const now = this.ctx.currentTime; const osc1 = this.ctx.createOscillator(); const gain1 = this.ctx.createGain(); osc1.type = 'sine'; osc1.frequency.setValueAtTime(freq, now);

const osc2 = this.ctx.createOscillator(); const gain2 = this.ctx.createGain(); osc2.type = 'sine'; osc2.frequency.setValueAtTime(freq * 2.76, now);

gain1.gain.setValueAtTime(0, now); gain1.gain.linearRampToValueAtTime(0.3, now + 0.02); gain1.gain.exponentialRampToValueAtTime(0.0001, now + duration);

gain2.gain.setValueAtTime(0, now); gain2.gain.linearRampToValueAtTime(0.1, now + 0.02); gain2.gain.exponentialRampToValueAtTime(0.0001, now + duration * 0.7);

osc1.connect(gain1); osc2.connect(gain2); gain1.connect(this.ctx.destination); gain2.connect(this.ctx.destination);

osc1.start(now); osc2.start(now); osc1.stop(now + duration); osc2.stop(now + duration); } catch { // ignore } }

// 敲击铜磬 (528Hz 纯正泛音) playSingingBowl(freq: number = 528, duration: number = 5.5) { if (!this.enabled) return; try { this.init();

```
if (!this.ctx) return;
const now = this.ctx.currentTime; // 触击声 const oscStrike = this.ctx.createOscillator(); const gainStrike = this.ctx.createGain(); oscStrike.type = 'triangle'; oscStrike.frequency.setValueAtTime(freq * 3.4, now); gainStrike.gain.setValueAtTime(0.18, now); gainStrike.gain.exponentialRampToValueAtTime(0.001, now + 0.08); oscStrike.connect(gainStrike); gainStrike.connect(this.ctx.destination); oscStrike.start(now); oscStrike.stop(now + 0.08);
// 主基频 const osc1 = this.ctx.createOscillator(); const gain1 = this.ctx.createGain(); osc1.type = 'sine'; osc1.frequency.setValueAtTime(freq, now);
// 泛音 const osc2 = this.ctx.createOscillator(); const gain2 = this.ctx.createGain(); osc2.type = 'sine'; osc2.frequency.setValueAtTime(freq * 2.76, now);
gain1.gain.setValueAtTime(0, now); gain1.gain.linearRampToValueAtTime(0.38, now + 0.03); gain1.gain.exponentialRampToValueAtTime(0.0001, now + duration);
gain2.gain.setValueAtTime(0, now); gain2.gain.linearRampToValueAtTime(0.16, now + 0.03); gain2.gain.exponentialRampToValueAtTime(0.0001, now + duration * 0.75);
osc1.connect(gain1); osc2.connect(gain2); gain1.connect(this.ctx.destination); gain2.connect(this.ctx.destination);
osc1.start(now); osc2.start(now); osc1.stop(now + duration); osc2.stop(now + duration); } catch { // ignore } }
// 瓷器碰杯
```

playPorcelainClink() { if (!this.enabled) return; try { this.init(); if (!this.ctx) return; const now = this.ctx.currentTime; const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'sine'; osc.frequency.setValueAtTime(2400, now); osc.frequency.exponentialRampToValueAtTime(1800, now + 0.08); gain.gain.setValueAtTime(0.15, now); gain.gain.exponentialRampToValueAtTime(0.001, now + 0.12); osc.connect(gain); gain.connect(this.ctx.destination); osc.start(now); osc.stop(now + 0.12); } catch { // ignore } }

// 点火火柴 playMatchStrike() { if (!this.enabled) return; try { this.init(); if (!this.ctx) return; const now = this.ctx.currentTime; const bufferSize = Math.floor(this.ctx.sampleRate * 0.15); const buffer = this.ctx.createBuffer(1, bufferSize, this.ctx.sampleRate); const data = buffer.getChannelData(0); for (let i = 0; i < bufferSize; i++) { data[i] = (Math.random() * 2 - 1) * Math.exp(-i / (bufferSize * 0.3)); }

const noise = this.ctx.createBufferSource(); noise.buffer = buffer; const filter = this.ctx.createBiquadFilter(); filter.type = 'bandpass'; filter.frequency.setValueAtTime(1800, now); filter.Q.setValueAtTime(3, now); const gain = this.ctx.createGain(); gain.gain.setValueAtTime(0.2, now); gain.gain.exponentialRampToValueAtTime(0.001, now + 0.15); noise.connect(filter); filter.connect(gain); gain.connect(this.ctx.destination); noise.start(now); } catch {

```
// ignore } }
// 摆木放置 playWoodPlacement() { if (!this.enabled) return; try { this.init(); if (!this.ctx) return; const now = this.ctx.currentTime; const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain(); osc.type = 'triangle'; osc.frequency.setValueAtTime(180, now); osc.frequency.exponentialRampToValueAtTime(60, now + 0.08); gain.gain.setValueAtTime(0.2, now); gain.gain.exponentialRampToValueAtTime(0.001, now + 0.08); osc.connect(gain); gain.connect(this.ctx.destination); osc.start(now); osc.stop(now + 0.08); } catch { // ignore } }
// 熄灭吹气 playExtinguish() { if (!this.enabled) return; try { this.init(); if (!this.ctx) return; const now = this.ctx.currentTime; const bufferSize = Math.floor(this.ctx.sampleRate * 0.12); const buffer = this.ctx.createBuffer(1, bufferSize, this.ctx.sampleRate); const data = buffer.getChannelData(0); for (let i = 0; i < bufferSize; i++) { data[i] = (Math.random() * 2 - 1) * Math.exp(-i / (bufferSize * 0.2)); } const noise = this.ctx.createBufferSource(); noise.buffer = buffer; const filter = this.ctx.createBiquadFilter(); filter.type = 'lowpass'; filter.frequency.setValueAtTime(800, now); const gain = this.ctx.createGain(); gain.gain.setValueAtTime(0.15, now); gain.gain.exponentialRampToValueAtTime(0.001, now + 0.12); noise.connect(filter); filter.connect(gain);
```

gain.connect(this.ctx.destination); noise.start(now); } catch { // ignore } }

// 清脆风铃轻吟 playWindChimes() { if (!this.enabled) return; try { this.init(); if (!this.ctx) return; const now = this.ctx.currentTime; // 五度/五声空灵风铃频率 (E6, G6, A6, B6, E7) const freqs = [1318.5, 1567.98, 1760.0, 1975.53, 2637.0]; freqs.forEach((freq, idx) => { if (!this.ctx) return; const delay = idx * 0.045 + Math.random() * 0.02; const noteStart = now + delay; const osc = this.ctx.createOscillator(); const gain = this.ctx.createGain();

osc.type = 'sine'; osc.frequency.setValueAtTime(freq, noteStart); // 微小微颤 osc.frequency.exponentialRampToValueAtTime(freq * 0.996, noteStart + 0.8);

gain.gain.setValueAtTime(0.0001, noteStart); gain.gain.linearRampToValueAtTime(0.08, noteStart + 0.015); gain.gain.exponentialRampToValueAtTime(0.0001, noteStart + 0.85);

osc.connect(gain); gain.connect(this.ctx.destination);

osc.start(noteStart); osc.stop(noteStart + 0.9); }); } catch { // ignore } } }

export const sound = new SoundEngine();

/* ===== 源文件：src/utils/tablePlan.ts ===== */ /**

- @file tablePlan.ts
- @description 供案桌面平面布局求解器。

*

- 坐标系与 `SceneItem.position` 完全一致：
- - x：相对案心的横向偏移，负为左、正为右；
- - y：案面纵深，0 = 紧贴台面前沿，负值向案里（后方）退，可用带约 [-54, 0]。

*

- 新贡品上案时不再使用「按数量奇偶取固定偏移」的粗放排布（多件时会层层重叠），
- 而是：1) 汇总案面既有占位；2) 在案面可用平面内采样候选点，用椭圆距离判定干涉
- （横向按占地半径严格要求，纵向按透视厚度放宽，允许前后错落遮挡）；
- 3) 按「不重叠 / 纵深合宜 / 近案心 / 左右均衡」评分选出相对最优摆位；
- 4) 案面排满时自动降级为「干涉最小」的点，保证永远有解。

*/ import { SceneItem } from '../types';

export interface PlanSpot { x: number; y: number; }

/** 案面可摆放平面范围（x 横向，y 纵深，已收敛掉案沿与后幕） */ export const TABLE_PLAN_BOUNDS = { xMin: -240, xMax: 240, yMin: -50, yMax: -4, };

/**

- 未记录 position 的固定陈设默认摆位。
- 与 PropsLayer 的默认布局基准值保持同量纲，用于占位避让。

*/ const DEFAULT_SPOTS: Record<string, PlanSpot> = { // 与 PropsLayer 默认布局同基准（xOffset 已随桌宽放大，取常用桌宽下的近似值） vase_left: { x: -180, y: -46 }, vase_right: { x: 180, y: -46 }, candle_left: { x: -82, y: -40 }, candle_right: { x: 82, y: -40 }, fruit_left: { x: -131, y: -28 }, fruit_right: { x: 131, y: -28 }, censer: { x: 0, y: -38 }, tea_cups: { x: 0, y: -18 }, singing_bowl: { x: -69, y: -8 }, };

/**

- 占地半宽（横向）与透视厚度（纵深）。
- 横向按道具实际占地半宽换算到布局量纲；纵深因台面进深仅约 54 单位而显著压缩，
- 取较小的厚度值，允许前后错落（后排被前排部分遮挡是透视中的自然观感）。

*/ const FOOTPRINT = { vase: { rx: 68, ry: 16 },

candle: { rx: 13, ry: 8 }, censer: { rx: 39, ry: 12 }, tea_cups: { rx: 49, ry: 12 }, singing_bowl: { rx: 37, ry: 10 }, fruit: { rx: 49, ry: 14 }, flower: { rx: 59, ry: 14 }, } as const;

const DEFAULT_FOOTPRINT = { rx: 46, ry: 12 };

interface Occupancy extends PlanSpot { rx: number; ry: number; }

interface Candidate extends PlanSpot { /** 归一化干涉余量：≥0 表示与既有道具不重叠，负值代表越挤 */ clearance: number; }

export interface TablePlanOptions { /** 待摆放道具的占地半宽，默认取花束尺寸 */ radius?: number; /** 待摆放道具的纵深厚度，默认取花束尺寸 */ depthRadius?: number; /** 期望的最小横向净间隙，默认 12；案面拥挤时自动放宽 */ minGap?: number; /** 期望的纵深位置（越大越靠案前），默认 -16 */ preferredDepth?: number; }

const footprintOf = (id: string, item: SceneItem): { rx: number; ry: number } => { if (id.startsWith('vase')) return FOOTPRINT.vase; if (id.startsWith('candle')) return FOOTPRINT.candle; if (id === 'censer') return FOOTPRINT.censer; if (id === 'tea_cups') return FOOTPRINT.tea_cups; if (id === 'singing_bowl') return FOOTPRINT.singing_bowl; if (id.startsWith('fruit')) return FOOTPRINT.fruit; if (id.startsWith('offered_flower')) return FOOTPRINT.flower; switch (item.type) { case 'vase': return FOOTPRINT.vase; case 'candle': return FOOTPRINT.candle; case 'censer': return FOOTPRINT.censer; case 'tea_cups': return FOOTPRINT.tea_cups; case 'singing_bowl': return FOOTPRINT.singing_bowl;

case 'fruit': return FOOTPRINT.fruit; case 'flower': return FOOTPRINT.flower; default: return DEFAULT_FOOTPRINT; } };

const collectOccupancy = (items: Record<string, SceneItem | undefined>): Occupancy[] => { const list: Occupancy[] = []; for (const [id, item] of Object.entries(items)) { if (!item) continue; const spot = item.position ?? DEFAULT_SPOTS[id] ?? { x: 0, y: -18 }; const fp = footprintOf(id, item); list.push({ x: spot.x, y: spot.y, rx: fp.rx, ry: fp.ry }); } return list; };

const solve = (occupancy: Occupancy[], options: TablePlanOptions): PlanSpot => { const { radius = FOOTPRINT.flower.rx, depthRadius = FOOTPRINT.flower.ry, minGap = 12, preferredDepth = -16, } = options;

// 案面横向重心：把新贡品补到相对空旷的一侧，避免全堆一边 const weightSum = occupancy.reduce((s, o) => s + o.rx, 0) || 1; const centerOfMass = occupancy.reduce((s, o) => s + o.x * o.rx, 0) / weightSum; const lean = Math.sign(centerOfMass);

// 按占地半径收边，避免道具压在案沿上（留出半个身位的安全余量） const xLimit = Math.max(60, TABLE_PLAN_BOUNDS.xMax - radius * 0.8);

const candidates: Candidate[] = []; for (let y = TABLE_PLAN_BOUNDS.yMin; y <= TABLE_PLAN_BOUNDS.yMax; y += 3) { for (let x = -xLimit; x <= xLimit; x += 8) { let clearance = Infinity; for (const o of occupancy) { const kx = o.rx + radius + minGap; const ky = o.ry + depthRadius + minGap * 0.35; const nx = (x - o.x) / kx; const ny = (y - o.y) / ky; clearance = Math.min(clearance, Math.hypot(nx, ny) - 1); } candidates.push({ x, y, clearance }); } }

const scoreOf = (c: Candidate): number => { const clearanceScore = Math.max(-1, Math.min(c.clearance, 1)) * 3.2; const centeringScore = -(Math.abs(c.x) / TABLE_PLAN_BOUNDS.xMax) * 0.9; const depthScore = -(Math.abs(c.y - preferredDepth) / 54) * 1.6; const balanceScore = -(c.x / TABLE_PLAN_BOUNDS.xMax) * lean * 0.8; return clearanceScore + centeringScore + depthScore + balanceScore; };

// 逐级放宽干涉要求：先要求有净间隙，再退到「相切即可」，最后允许轻微重叠 for (const threshold of [0, -0.15, -0.35, -0.6]) { const feasible = candidates.filter((c) => c.clearance >= threshold); if (feasible.length) { return feasible.reduce((best, c) => (scoreOf(c) > scoreOf(best) ? c : best)); } }

// 案面已排满：取干涉最小的点 return candidates.reduce((best, c) => (scoreOf(c) > scoreOf(best) ? c : best)); };

/**

- 为一件新贡品寻找案面上相对最优的摆位。

*/ export const findOptimalTableSpot = (items: Record<string, SceneItem | undefined>, options: TablePlanOptions = {}): PlanSpot => { const occupancy = collectOccupancy(items); const spot = solve(occupancy, options); return { x: Math.round(spot.x), y: Math.round(spot.y) }; };

/**

- 批量布局：逐件求解，并把已排定的位置计入占用，避免一次性排布时互相重叠。

*/ export const planSpotsSequentially = (items: Record<string, SceneItem | undefined>, count: number, options: TablePlanOptions = {}): PlanSpot[] => { const occupancy = collectOccupancy(items); const rx = options.radius ?? FOOTPRINT.flower.rx; const ry = options.depthRadius ?? FOOTPRINT.flower.ry; const spots: PlanSpot[] = [];

for (let i = 0; i < count; i++) { const spot = solve(occupancy, options); occupancy.push({ x: spot.x, y: spot.y, rx, ry }); spots.push({ x: Math.round(spot.x), y: Math.round(spot.y) });

}

return spots; };

/* ===== 源文件：src/utils/timeOfDay.ts ===== */ /**

- @file timeOfDay.ts
- @description 依据本机系统时间推算自然天色时辰（供"时间跟随系统"模式使用）。

*/

import { TimeOfDay } from '../types';

/** 时段划分（本地时间，24h 制）：白昼 6:00 – 16:59 / 黄昏 17:00 – 18:59 / 夜色 19:00 – 5:59 */ export const SUNSET_START_HOUR = 17; export const NIGHT_START_HOUR = 19; export const DAY_START_HOUR = 6;

export const resolveSystemTimeOfDay = (date: Date = new Date()): TimeOfDay => { const h = date.getHours(); if (h >= SUNSET_START_HOUR && h < NIGHT_START_HOUR) return 'sunset'; if (h >= NIGHT_START_HOUR || h < DAY_START_HOUR) return 'night'; return 'day'; };

export const TIME_OF_DAY_LABEL: Record<TimeOfDay, string> = { day: '白昼晨辉', sunset: '暖调黄昏', night: '静谧微夜', };

/** 格式化为 HH:MM（本地时间） */ export const formatClock = (date: Date = new Date()): string => `${String(date.getHours()).padStart(2, '0')}:${String(date.getMinutes()).padStart(2, '0')}`;

/* ===== 源文件：src/utils/useWorldLoop.ts ===== */ /**

- @file useWorldLoop.ts
- @description 供案世界时钟与供品生命周期调度 Hook。

*/ import { useEffect, useRef, useState, useCallback } from 'react'; import { GameItemData, WorldContext, EntityEvent, LifecycleStage } from '../types/entity'; import { LightingSettings } from '../types';

interface UseWorldLoopProps { items: Record<string, GameItemData>; onUpdateItems: (updater: (prev: Record<string, GameItemData>) => Record<string, GameItemData>) => void; lighting: LightingSettings; soundEnabled: boolean; initialTimeScale?: number;

```
onEntityEvent?: (event: EntityEvent) => void; }

export function useWorldLoop({ items, onUpdateItems, lighting, soundEnabled, initialTimeScale = 1.0, onEntityEvent, }: UseWorldLoopProps) { const [timeScale, setTimeScale] = useState<number>(initialTimeScale); const [isPaused, setIsPaused] = useState<boolean>(false);

const lastTimeRef = useRef<number>(performance.now()); const lastCommitTimeRef = useRef<number>(performance.now()); const itemsRef = useRef(items); itemsRef.current = items;

const onEntityEventRef = useRef(onEntityEvent); onEntityEventRef.current = onEntityEvent;

useEffect(() => { let animId: number;

const tick = (now: number) => { const deltaMs = Math.min(100, now - lastTimeRef.current); lastTimeRef.current = now;

const deltaSec = deltaMs / 1000; const effectiveScale = isPaused ? 0 : timeScale;

if (effectiveScale > 0) { const commitInterval = effectiveScale <= 1 ? 1000 : effectiveScale <= 10 ? 300 : 150; const timeSinceLastCommit = now - lastCommitTimeRef.current; const shouldPeriodicCommit = timeSinceLastCommit >= commitInterval;

const currentItems = itemsRef.current; let hasStructuralChange = false; let hasValueChange = false; const nextItems: Record<string, GameItemData> = {}; const eventsToFire: EntityEvent[] = [];

for (const id of Object.keys(currentItems)) { const item = currentItems[id]; if (!item) continue;

// 仅对具有 lifecycle 的物品进行生命周期心跳推进 if (!item.lifecycle) { nextItems[id] = item;
```

```
continue; }

const lc = item.lifecycle; const isCandle = id.startsWith('candle'); const isCenser = id === 'censer';

// 未点燃的蜡烛或香炉暂停衰减 if ((isCandle || isCenser) && !item.isLit) { nextItems[id] = item; continue; }

const nextAge = lc.age + deltaSec * (lc.decayRate || 1.0) * effectiveScale; const progress = Math.min(1.0, nextAge / Math.max(1, lc.duration || 180)); const health = Math.max(0, 1.0 - progress);

let stage: LifecycleStage = lc.stage; let isLit = item.isLit;

if (isCandle) { if (progress < 0.40) stage = 'blooming'; else if (progress < 0.70) stage = 'wilting'; else if (progress < 0.96) stage = 'withered'; else { stage = 'expired'; isLit = false; } } else { if (progress < 0.15) stage = 'budding'; else if (progress < 0.65) stage = 'blooming'; else if (progress < 0.90) { if (stage === 'blooming') { eventsToFire.push({ type: 'petal_drop', entityId: id }); } stage = 'wilting'; } else if (progress < 1.0) stage = 'withered'; else { stage = 'expired'; if (lc.autoDespawn) { eventsToFire.push({ type: 'despawn', entityId: id }); hasStructuralChange = true; continue; // 移除已消亡物品 } } }

const updated: GameItemData = { ...item, isLit,
```

```
lifecycle: { ...lc, age: nextAge, health, stage, }, };
if (stage !== lc.stage || isLit !== item.isLit) { hasStructuralChange = true; } else { hasValueChange = true; }
nextItems[id] = updated; }
if (hasStructuralChange || (hasValueChange && shouldPeriodicCommit)) { lastCommitTimeRef.current = now; onUpdateItems(() => nextItems); }
if (eventsToFire.length > 0 && onEntityEventRef.current) { eventsToFire.forEach((ev) => onEntityEventRef.current?.(ev)); } }
animId = requestAnimationFrame(tick); };
lastTimeRef.current = performance.now(); lastCommitTimeRef.current = performance.now(); animId = requestAnimationFrame(tick);
return () => { cancelAnimationFrame(animId); }; }, [isPaused, timeScale, lighting, soundEnabled, onUpdateItems]);
const togglePause = useCallback(() => { setIsPaused((p) => !p); }, []);
return { timeScale, setTimeScale, isPaused, togglePause, }; }
```